

AE_544_LecNote06_Variational_Methods_Ch06

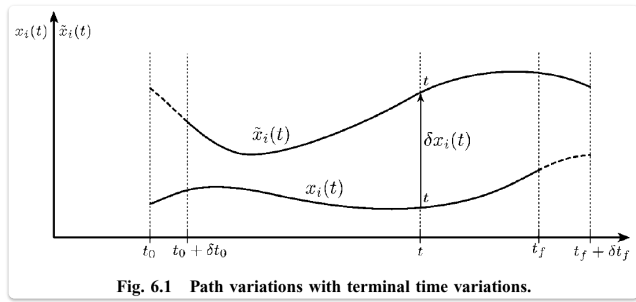
Disclaimers

This repository contains markdown-based lecture notes for the Spring 2025 semester. These materials are intended solely for students enrolled in the course during this term. Mistakes may exist and will be corrected as they are identified, but the official textbook remains the ultimate reference for accuracy.

In this chapter, we begin by developing basic concepts from variational calculus and then turn to the development of the most important results in variational mechanics:

1. A family of variational principles due to Hamilton that hold for the motion of very general systems, including distributed parameter systems.
2. Hamilton's principal function $\mathcal{S} \equiv \int_{t_0}^{t_f} \mathcal{L} dt$, which has several important properties.
3. Extensions of Lagrange's equations for the case of distributed parameter systems.

Fundamentals of Variational Calculus



We seek to determine a space time trajectory or path $\mathbf{x} \in \mathbb{R}^n$ that causes a given functional $\mathcal{J}(\mathbf{x}(t), t_0, t_f)$ to achieve a local minimum (or maximum). $\mathcal{J}$ is considered a functional because its argument list contains a vector of unknown functions $\mathbf{x}(t) = (x_1(t), x_2(t), \dots, x_n(t))^T$.

Let us consider the special case that $\mathcal{J}(\mathbf{x}(t), t_0, t_f)$ is expressible as a path integral:

$$\mathcal{J} = \mathcal{J}(\mathbf{x}(t), t_0, t_f) = \int_{t_0}^{t_f} \mathcal{F}(\mathbf{x}(t), \dot{\mathbf{x}}(t), t) dt \quad (6.1)$$

Suppose $\mathcal{F}$ and $\mathbf{x}$ are continuous functions and twice differentiable with respect to all argument (functions of class C_2), and $\mathbf{x}(t)$ is the minimal solution from t_0 to t_f . Denote a neighboring path and times as $\{\tilde{\mathbf{x}}(t), \tilde{t}_0, \tilde{t}_f\}$, then the first path variation is

$$\delta \mathbf{x}(t) \equiv \tilde{\mathbf{x}}(t) - \mathbf{x}(t) \quad (6.2)$$

$$\delta \dot{\mathbf{x}}(t) = \dot{\tilde{\mathbf{x}}}(t) - \dot{\mathbf{x}}(t) \equiv \frac{d}{dt}(\delta \mathbf{x}) \quad (6.3)$$

Thus the variational operator $\delta(\cdot)$ and the differential operator $d/dt(\cdot)$ are interchangeable.

The varied state vector and its time derivative are

$$\begin{aligned} \tilde{\mathbf{x}}(t) &= \mathbf{x}(t) + \delta \mathbf{x}(t) \\ \dot{\tilde{\mathbf{x}}}(t) &= \dot{\mathbf{x}}(t) + \delta \dot{\mathbf{x}}(t) \end{aligned} \quad (6.5)$$

The varied path integral of $\tilde{\mathcal{J}}$ is

$$\tilde{\mathcal{J}} = \mathcal{J}(\tilde{\mathbf{x}}(t), \tilde{t}_0, \tilde{t}_f) = \int_{\tilde{t}_0}^{\tilde{t}_f} \mathcal{F}(\mathbf{x}(t) + \delta \mathbf{x}(t), \dot{\mathbf{x}}(t) + \delta \dot{\mathbf{x}}(t), t) dt \quad (6.6)$$

The variation $\delta \mathcal{J}$ is obtained as

$$\begin{aligned} \delta \mathcal{J} &\equiv \tilde{\mathcal{J}} - \mathcal{J} \\ &= \int_{t_0 + \delta t_0}^{t_f + \delta t_f} \mathcal{F}(\mathbf{x}(t) + \delta \mathbf{x}(t), \dot{\mathbf{x}}(t) + \delta \dot{\mathbf{x}}(t), t) dt - \int_{t_0}^{t_f} \mathcal{F}(\mathbf{x}(t), \dot{\mathbf{x}}(t), t) dt \end{aligned} \quad (6.7)$$

Express the first integral using Taylor series, and truncate at the first-order terms, we have

$$\begin{aligned}
& \int_{t_0+\delta t_0}^{t_f+\delta t_f} \mathcal{F}(\mathbf{x}(t) + \delta \mathbf{x}(t), \dot{\mathbf{x}}(t) + \delta \dot{\mathbf{x}}(t), t) dt \\
& \approx \int_{t_0}^{t_f} \mathcal{F}(\mathbf{x}(t), \dot{\mathbf{x}}(t), t) dt + \left[\frac{\partial}{\partial \mathbf{x}} \left(\int_{t_0}^{t_f} \mathcal{F}(\mathbf{x}(t), \dot{\mathbf{x}}(t), t) dt \right) \right]^T \delta \mathbf{x} \\
& + \left[\frac{\partial}{\partial \dot{\mathbf{x}}} \left(\int_{t_0}^{t_f} \mathcal{F}(\mathbf{x}(t), \dot{\mathbf{x}}(t), t) dt \right) \right]^T \delta \dot{\mathbf{x}} + \frac{\partial}{\partial t_0} \left(\int_{t_0}^{t_f} \mathcal{F}(\mathbf{x}(t), \dot{\mathbf{x}}(t), t) dt \right) \delta t_0 \\
& + \frac{\partial}{\partial t_f} \left(\int_{t_0}^{t_f} \mathcal{F}(\mathbf{x}(t), \dot{\mathbf{x}}(t), t) dt \right) \delta t_f \\
& \equiv \int_{t_0}^{t_f} \mathcal{F}(\mathbf{x}(t), \dot{\mathbf{x}}(t), t) dt + \int_{t_0}^{t_f} \left(\frac{\partial}{\partial \mathbf{x}} \mathcal{F}(\mathbf{x}(t), \dot{\mathbf{x}}(t), t) \cdot \delta \mathbf{x} \right) dt \\
& + \int_{t_0}^{t_f} \left(\frac{\partial}{\partial \dot{\mathbf{x}}} \mathcal{F}(\mathbf{x}(t), \dot{\mathbf{x}}(t), t) \cdot \delta \dot{\mathbf{x}} \right) dt + \left[-\mathcal{F}(\mathbf{x}(t_0), \dot{\mathbf{x}}(t_0), t_0) \cdot \delta t_0 \right] \\
& + \mathcal{F}(\mathbf{x}(t_f), \dot{\mathbf{x}}(t_f), t_f) \cdot \delta t_f
\end{aligned} \tag{1st term in 6.7}$$

① Taylor expansion for multi-variable function $K(\mathbf{x}, \dot{\mathbf{x}}, t_0, t_f)$ is

$$K = \left(\frac{\partial K}{\partial \mathbf{x}} \right)^T d\mathbf{x} + \left(\frac{\partial K}{\partial \dot{\mathbf{x}}} \right)^T d\dot{\mathbf{x}} + \frac{\partial K}{\partial t_0} dt_0 + \frac{\partial K}{\partial t_f} dt_f + O(\|d\mathbf{x}, d\dot{\mathbf{x}}, dt_0, dt_f\|^2)$$

Plug the 1st order Taylor expansion back into Eq. (6.7) and we have

$$\begin{aligned}
\delta \mathcal{J} = & \int_{t_0}^{t_f} \left[\left(\frac{\partial \mathcal{F}(\ast)}{\partial \mathbf{x}(t)} \right)^T \delta \mathbf{x}(t) \right] dt + \int_{t_0}^{t_f} \left[\left(\frac{\partial \mathcal{F}(\ast)}{\partial \dot{\mathbf{x}}(t)} \right)^T \delta \dot{\mathbf{x}}(t) \right] dt \\
& + \mathcal{F}(\mathbf{x}(t_f), \dot{\mathbf{x}}(t_f), t_f) \delta t_f - \mathcal{F}(\mathbf{x}(t_0), \dot{\mathbf{x}}(t_0), t_0) \delta t_0
\end{aligned} \tag{6.8}$$

If $\mathcal{J}$ is minimized, a necessary condition is that for arbitrary $\delta \mathbf{x}$, δt_0 , δt_f , the variation $\delta \mathcal{J}$ has to vanish, or in other words, be equal to 0.

Notice that we can do integral by parts for the second term in Eq. (6.8), which is

$$\int_{t_0}^{t_f} \left[\frac{\partial \mathcal{F}(\ast)}{\partial \dot{\mathbf{x}}(t)} \right]^T \delta \dot{\mathbf{x}}(t) dt = \left(\left[\frac{\partial \mathcal{F}(\ast)}{\partial \dot{\mathbf{x}}(t)} \right]^T \delta \mathbf{x}(t) \right) \Big|_{t_0}^{t_f} - \int_{t_0}^{t_f} \frac{d}{dt} \left[\frac{\partial \mathcal{F}(\ast)}{\partial \dot{\mathbf{x}}(t)} \right]^T \delta \mathbf{x}(t) dt \tag{6.10}$$

This finally gives us an expression of the variation as a function of just $\delta \mathbf{x}$ and terminal times t_0 and t_f , which is

$$\begin{aligned}
\delta \mathcal{J} = & \int_{t_0}^{t_f} \left[\left(\frac{\partial \mathcal{F}(\ast)}{\partial \mathbf{x}(t)} \right)^T \delta \mathbf{x}(t) \right] dt + \left(\left[\frac{\partial \mathcal{F}(\ast)}{\partial \dot{\mathbf{x}}(t)} \right]^T \delta \mathbf{x}(t) \right) \Big|_{t_0}^{t_f} - \int_{t_0}^{t_f} \frac{d}{dt} \left[\frac{\partial \mathcal{F}(\ast)}{\partial \dot{\mathbf{x}}(t)} \right]^T \delta \mathbf{x}(t) dt \\
& + \mathcal{F}(\mathbf{x}(t_f), \dot{\mathbf{x}}(t_f), t_f) \delta t_f - \mathcal{F}(\mathbf{x}(t_0), \dot{\mathbf{x}}(t_0), t_0) \delta t_0 \\
= & \int_{t_0}^{t_f} \left[\frac{\partial \mathcal{F}(\ast)}{\partial \mathbf{x}(t)} - \frac{d}{dt} \left(\frac{\partial \mathcal{F}(\ast)}{\partial \dot{\mathbf{x}}(t)} \right) \right]^T \delta \mathbf{x}(t) dt + \left(\left[\frac{\partial \mathcal{F}(\ast)}{\partial \dot{\mathbf{x}}(t)} \right]^T \delta \mathbf{x}(t) \right) \Big|_{t_0}^{t_f} \\
& + \mathcal{F}(\mathbf{x}(t_f), \dot{\mathbf{x}}(t_f), t_f) \delta t_f - \mathcal{F}(\mathbf{x}(t_0), \dot{\mathbf{x}}(t_0), t_0) \delta t_0
\end{aligned} \tag{6.11}$$

① The reason we prefer Eq. (6.11) over Eq. (6.8) is that usually we have constraints on the initial and final state $\mathbf{x}(t_0)$ or $\mathbf{x}(t_f)$ instead of their derivatives.

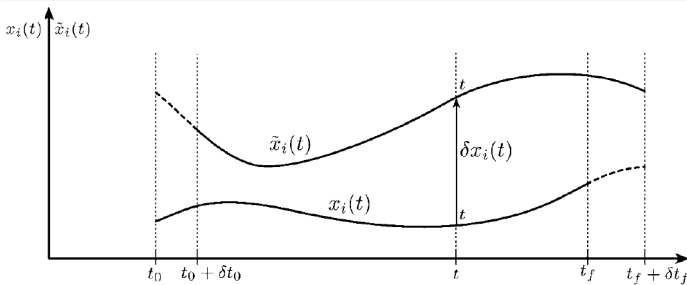


Fig. 6.1 Path variations with terminal time variations.

Notice that $\delta \mathbf{x}(t)$, $\delta \mathbf{x}(t_0)$, $\delta \mathbf{x}(t_f)$, δt_0 , δt_f are all independent and shall be considered independently. This leads to the necessary conditions below:

Euler-Lagrange equations:

$$\frac{\partial \mathcal{F}(\ast)}{\partial \mathbf{x}(t)} - \frac{d}{dt} \left(\frac{\partial \mathcal{F}(\ast)}{\partial \dot{\mathbf{x}}(t)} \right) = 0 \quad (6.12)$$

and transversality conditions:

$$\begin{aligned} \left[\frac{\partial \mathcal{F}(\mathbf{x}(t_0), \dot{\mathbf{x}}(t_0), t_0)}{\partial \dot{\mathbf{x}}(t_0)} \right]^T \cdot \delta \mathbf{x}(t_0) &= 0 \\ \left[\frac{\partial \mathcal{F}(\mathbf{x}(t_f), \dot{\mathbf{x}}(t_f), t_f)}{\partial \dot{\mathbf{x}}(t_f)} \right]^T \cdot \delta \mathbf{x}(t_f) &= 0 \\ \mathcal{F}(\mathbf{x}(t_f), \dot{\mathbf{x}}(t_f), t_f) \cdot \delta t_f &= 0 \\ \mathcal{F}(\mathbf{x}(t_0), \dot{\mathbf{x}}(t_0), t_0) \cdot \delta t_0 &= 0 \end{aligned} \quad (6.13)$$

⚠ The fact they are independent does NOT necessarily imply that they can be chosen arbitrarily.

They still depends on the available initial conditions or terminal constraints. (Distinguish the variation from the virtual displacements in d'Alembert's Principle.)

An important special case is the "fixed endpoint problem" in which the boundary conditions $\delta \mathbf{x}(t), \mathbf{x}(t_0), \mathbf{x}(t_f), t_0, t_f$ are all given and fixed, rendering the variations $\delta \mathbf{x}(t_0), \delta \mathbf{x}(t_f), \delta t_0, \delta t_f$ all zeros. In this case, all the transversality conditions are satisfied naturally, and the Euler-Lagrange equations are the only necessary conditions for an extreme $\mathcal{J}$.

Hamilton's Principal Function $\mathcal{S}$

Recall the Lagrange's Equations with all potential forces:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j} = 0 \quad (5.153)$$

which is exactly in the form of Euler-Lagrange equation we just obtained:

$$\frac{\partial \mathcal{F}(\ast)}{\partial \mathbf{x}(t)} - \frac{d}{dt} \left(\frac{\partial \mathcal{F}(\ast)}{\partial \dot{\mathbf{x}}(t)} \right) = 0 \quad (6.12)$$

Therefore, we can conclude that the Hamilton's principal function $\mathcal{S}$

$$\mathcal{S} = \int_{t_0}^{t_f} \mathcal{L} dt \quad (6.16)$$

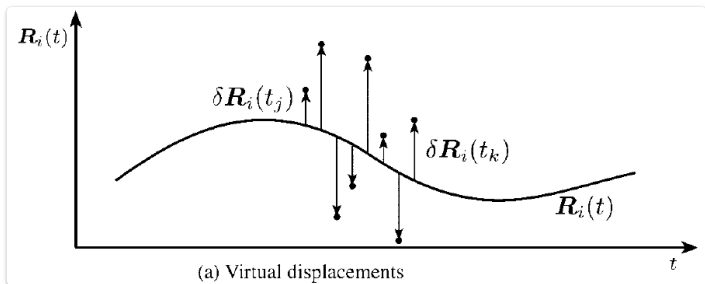
is an extremum for dynamical motions with zero generalized forces on the right side of Eq. (6.15), and for this class of systems we have the simplest version of Hamilton's variational principle

$$\delta \mathcal{S} = 0. \quad (6.17)$$

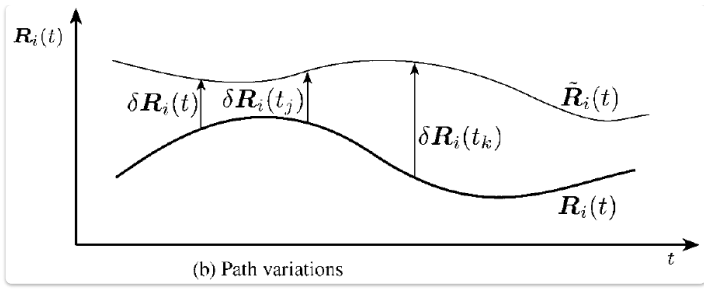
Hamilton's Variational Principles

Virtual Displacements vs Path Variations

Consider a "true trajectory" (one satisfying Newton's laws or Lagrange's equations) $\mathbf{R}_i(t)$.



At each give epoch t_j , there is an arbitrary virtual displacement $\delta \mathbf{R}_i(t_j)$, so the entire arbitrary differential virtual displacement locates an infinite set of neighboring points. The virtual displacements $\delta \mathbf{R}_i(t_j)$ and $\delta \mathbf{R}_i(t_k)$ bear no relationship to each other; they are arbitrary and independent as long as they are of differential magnitude.



Differently, in Fig. 6.2b, the arbitrary variational path $\tilde{\mathbf{R}}_i(t)$ is another C_2 smooth path nearly the true trajectory, and the variation $\delta \mathbf{R}_i(t) = \tilde{\mathbf{R}}_i(t) - \mathbf{R}_i(t)$ is also a C_2 function. So, two displacements $\delta \mathbf{R}_i(t_j)$ and $\delta \mathbf{R}_i(t_k)$ at different epochs are dependent. If t_j and t_k are close enough, this indicates they must satisfy the C_2 smooth condition.

Both $\mathbf{R}_i(t)$ and $\tilde{\mathbf{R}}_i(t)$ satisfy all system constraints, but only $\mathbf{R}_i(t)$ are true paths satisfying the system equations of motion.

As a consequence of the fact that the general virtual displacements contain path variations as a special case, variational results obtained using virtual displacements typically hold for the special case of path variations as well.

✓ Virtual displacement $\supset$ Path variation

Hamilton's Principle Derived from D'Alembert's Principle

Let $\mathbf{F}_i$ include all virtually working forces, recall that the d'Alembert's principle is obtained by combining

$$\delta W_i \equiv \mathbf{F}_i \cdot \delta \mathbf{R}_i \quad (5.10)$$

and

$$\delta W - \sum_{i=1}^N m_i \ddot{\mathbf{R}}_i \cdot \delta \mathbf{R}_i = 0 \quad (5.19)$$

such that we have

$$\sum_{i=1}^N (\mathbf{F}_i - m_i \ddot{\mathbf{R}}_i) \cdot \delta \mathbf{R}_i = 0 \quad (6.18)$$

② Relation to other versions of d'Alembert's principle?

✓ Different forms of d'Alembert's principle:

1. (classical form, Eq. 5.24) $\sum_{i=1}^N m_i \ddot{\mathbf{R}}_i \cdot \frac{\partial \mathbf{R}_i}{\partial \mathbf{q}_j} = Q_j$
2. $\sum_{i=1}^N m_i \ddot{\mathbf{R}}_i \cdot \mathbf{v}_{ij} = Q_j$
3. (virtual power form, Eq. 5.30) $\sum_{i=1}^N [\mathbf{f}_i - m_i \dot{\mathbf{V}}_i] \cdot \mathbf{v}_{ij} = 0$

where $Q_j = \sum_{i=1}^N \mathbf{f}_i \cdot \frac{\partial \mathbf{R}_i}{\partial \mathbf{q}_j} = \sum_{i=1}^N \mathbf{f}_i \cdot \mathbf{v}_{ij}$ and $\mathbf{v}_{ij} = \frac{\partial \dot{\mathbf{R}}_i}{\partial \dot{\mathbf{q}}_j} = \frac{\partial \mathbf{R}_i}{\partial \mathbf{q}_j}$.

Although d'Alembert's principle was derived for the most general case that $\delta \mathbf{R}_i$ are instantaneous virtual displacements, this condition also holds for the special case that $\delta \mathbf{R}_i(t)$ are path variations with $\tilde{\mathbf{R}}_i(t) = \mathbf{R}_i(t) + \delta \mathbf{R}_i(t)$ locating an infinite family of neighboring varied paths.

The total kinetic energy definition of a system of N particles along the true path $\mathbf{R}_i(t)$ is

$$T = \frac{1}{2} \sum_{i=1}^N m_i \dot{\mathbf{R}}_i \cdot \dot{\mathbf{R}}_i \quad (6.19)$$

and the total kinetic energy function evaluated at the same moment but on the differentially displaced varied motion $\tilde{\mathbf{R}}_i(t)$ is

$$\tilde{T} = \frac{1}{2} \sum_{i=1}^N m_i \dot{\tilde{\mathbf{R}}}_i \cdot \dot{\tilde{\mathbf{R}}}_i \quad (6.20)$$

Thus, we have the path variation of kinetic energy as

$$\begin{aligned}
\delta T &= \tilde{T} - T \\
&= \frac{1}{2} \sum_{i=1}^N m_i \dot{\mathbf{R}}_i \cdot \dot{\mathbf{R}}_i - \frac{1}{2} \sum_{i=1}^N m_i \dot{\tilde{\mathbf{R}}}_i \cdot \dot{\tilde{\mathbf{R}}}_i \\
&= \frac{1}{2} \sum_{i=1}^N m_i \dot{\mathbf{R}}_i \cdot \dot{\mathbf{R}}_i - \frac{1}{2} \sum_{i=1}^N m_i (\dot{\mathbf{R}}_i + \delta \dot{\mathbf{R}}_i) \cdot (\dot{\mathbf{R}}_i + \delta \dot{\mathbf{R}}_i) \\
&\approx \sum_{i=1}^N m_i \dot{\mathbf{R}}_i \cdot \delta \dot{\mathbf{R}}_i
\end{aligned} \tag{6.21}$$

We can get the following relationship by applying the chain rule of the time derivative to a special summation :

$$\frac{d}{dt} \left(\sum_{i=1}^N m_i \dot{\mathbf{R}}_i \cdot \delta \mathbf{R}_i \right) = \sum_{i=1}^N m_i \ddot{\mathbf{R}}_i \cdot \delta \mathbf{R}_i + \sum_{i=1}^N m_i \dot{\mathbf{R}}_i \cdot \delta \dot{\mathbf{R}}_i \tag{6.22}$$

Notice that we have switch the order of derivatives and variations to get the above expression. The first term on RHS is the virtual work δW , and the second term on RHS is the just defined variation of kinetic energy, δT .

Therefore, what we get is actually

$$\frac{d}{dt} \left(\sum_{i=1}^N m_i \dot{\mathbf{R}}_i \cdot \delta \mathbf{R}_i \right) = \delta W + \delta T \tag{6.24}$$

Integrate the above equation and get the **most general form of Hamilton's principles**:

$$\int_{t_0}^{t_f} (\delta W + \delta T) dt = \left(\sum_{i=1}^N m_i \dot{\mathbf{R}}_i \cdot \delta \mathbf{R}_i \right) \Big|_{t_0}^{t_f} \tag{6.25}$$

It holds for the case of 1) arbitrary forces, 2) general constraints, and 3) general initial and final boundary conditions.

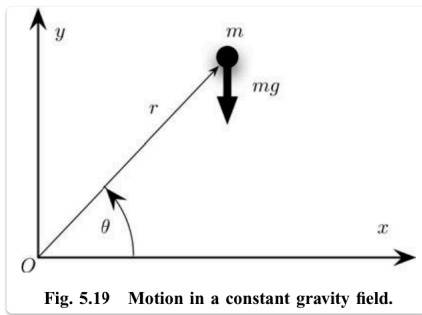
If the virtually working forces are conservative, then $\delta W = -\delta V$, and if attention is restricted to fixed endpoint problems, $\delta \mathbf{R}_i(t_0) = \delta \mathbf{R}_i(t_f) = 0$, then the the above general form specializes to

$$\int_{t_0}^{t_f} (\delta T - \delta V) dt = \int_{t_0}^{t_f} \delta(T - V) dt = \int_{t_0}^{t_f} \delta \mathcal{L} dt = \delta \int_{t_0}^{t_f} \mathcal{L} dt = 0 \tag{6.26}$$

Note that the interchangeability of variation operator and integral is generally true under holonomic constraints.

Example 6.5: motion in a constant gravity field

This example shows that Hamilton's principle of varying action can provide a direct path to determine the system motion that does not "pass through" Lagrange's equations.



(Example 5.18) Lagrange's equations:

$$\mathcal{L} = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2) - mgy$$

$$\frac{\partial \mathcal{L}}{\partial x} = 0, \quad \frac{\partial \mathcal{L}}{\partial y} = -mg$$

$$\frac{\partial \mathcal{L}}{\partial \dot{x}} = m\dot{x}, \quad \frac{\partial \mathcal{L}}{\partial \dot{y}} = m\dot{y}$$

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}} - \frac{\partial \mathcal{L}}{\partial x} = m\ddot{x} = 0$$

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{y}} - \frac{\partial \mathcal{L}}{\partial y} = m\ddot{y} + mg = 0$$

So, $\dot{x} = \dot{x}_0 = \text{constant}$, $y = y_0 + \dot{y}_0 t - \frac{1}{2}gt^2$.

Hamilton's principle: (Let's look at just vertical motion for simplicity.)

$$T = \frac{1}{2}m\dot{y}^2, V = mgy$$

$$\delta T = m\dot{y}\delta\dot{y}, \delta V = -\delta W = mg\delta y$$

$$\frac{\delta T}{\delta \dot{y}} = m\dot{y}$$

Recall

$$\int_{t_0}^{t_f} (\delta W + \delta T) dt = \left(\sum_{i=1}^N m_i \dot{\mathbf{R}}_i \cdot \delta \mathbf{R}_i \right) \Big|_{t_0}^{t_f} \quad (6.25)$$

Since $\delta y(0) = 0$ and $\delta y(f) = 0$,

$$\int_{t_0}^{t_f} (mg\delta y + m\dot{y}\delta\dot{y}) dt = 0$$

Simplified to:

$$\int_{t_0}^{t_f} (g\delta y + \dot{y}\delta\dot{y}) dt = 0$$

Since dt is arbitrary, there must be $g\delta y + \dot{y}\delta\dot{y} = 0$

Assume the solution has the form of infinite power series:

$$y(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + \dots$$

$$\dot{y}(t) = a_1 + 2a_2 t + 3a_3 t^2 + \dots$$

Then the variations are:

$$\delta y(t) = \delta a_0 + \delta a_1 t + \delta a_2 t^2 + \dots$$

$$\delta \dot{y}(t) = \delta a_1 + 2\delta a_2 t + 3\delta a_3 t^2 + \dots$$

(Notice this is variations of functions, so no δt appears.)

Plug in

$$\begin{aligned} & \{(2a_2 + g)t + 3a_3 t^2 + \dots\} \delta a_0 \\ & + \left\{ \left(a_2 + \frac{g}{2} \right) t^2 + 2a_3 t^3 + \dots \right\} \delta a_1 \\ & + \left\{ \frac{1}{3} (2a_2 + g) t^3 + \frac{3}{2} a_3 t^4 + \dots \right\} \delta a_2 \\ & + \dots = 0 \end{aligned}$$

Again since t is arbitrary, all coefficients must vanish, thus $a_2 = -\frac{1}{2}g$, and $a_3 = a_4 = \dots = 0$

So, the solution is $y(t) = a_0 + a_1 t - \frac{1}{2}gt^2$

🔗 Can a_0 and a_1 be determined? How?

Left as a after-class practice. Answer is in the textbook.

Apply Hamilton's Principle to Distributed Parameter Systems

We will mainly show two things:

1. Hamilton's principle can provide natural boundary conditions "automatically" for distributed parameter systems.
2. Hamilton's principle can be used to derive very general versions of Lagrange's equations that are the partial differential equations governing families of distributed parameter systems.

Hamilton's law of varying action in Eq. (6.25) provides an approach to directly solve for the system motion, without first deriving a differential equation.

Goal: Before we can apply Hamilton's principle to a distributed parameter system, we need to first extend it to a continuous body instead of relying on a system of particles.

Recall the Hamilton's principle:

$$\int_{t_0}^{t_f} (\delta W + \delta T) dt = \left(\sum_{i=1}^N m_i \dot{\mathbf{R}}_i \cdot \delta \mathbf{R}_i \right) \Big|_{t_0}^{t_f} \quad (6.25)$$

Denote the RHS as

$$\mathcal{B} = \sum_{i=1}^N m_i \dot{\mathbf{R}}_i \cdot \delta \mathbf{R}_i \quad (6.40)$$

Since we have used generalized coordinates for $\mathbf{R}_i(q_1, q_2, \dots, q_n, t)$, we can use partial derivatives to get an express for $\delta \mathbf{R}_i$, which leads to:

$$\mathcal{B} = \sum_{i=1}^N m_i \dot{\mathbf{R}}_i \cdot \delta \mathbf{R}_i = \sum_{i=1}^N m_i \dot{\mathbf{R}}_i \cdot \left(\sum_{j=1}^n \frac{\partial \mathbf{R}_i}{\partial q_j} \delta q_j \right)$$

Using the two tricks of 1) cancellation of dots and 2) switching the order of summations, we can connect $\mathcal{B}$ to kinetic energy T through:

$$\begin{aligned} \mathcal{B} &= \sum_{j=1}^n \left(\sum_{i=1}^N m_i \dot{\mathbf{R}}_i \cdot \frac{\partial \mathbf{R}_i}{\partial q_j} \right) \delta q_j \\ &= \sum_{j=1}^n \left(\sum_{i=1}^N m_i \dot{\mathbf{R}}_i \cdot \frac{\partial \dot{\mathbf{R}}_i}{\partial \dot{q}_j} \right) \delta q_j \\ &= \sum_{j=1}^n \left(\sum_{i=1}^N \frac{\partial}{\partial \dot{q}_j} \left(\frac{1}{2} m_i \dot{\mathbf{R}}_i^2 \right) \right) \delta q_j \\ &= \sum_{j=1}^n \frac{\partial T}{\partial \dot{q}_j} \delta q_j \end{aligned} \quad (6.43)$$

Substitute this back to the Hamilton's principle in Eq. (6.25)

$$\int_{t_0}^{t_f} (\delta W + \delta T) dt = \left(\sum_{i=1}^N m_i \dot{\mathbf{R}}_i \cdot \delta \mathbf{R}_i \right) \Big|_{t_0}^{t_f} \quad (6.25)$$

and we have

$$\int_{t_0}^{t_f} (\delta W + \delta T) dt = \left(\sum_{j=1}^n \frac{\partial T}{\partial \dot{q}_j} \delta q_j \right) \Big|_{t_0}^{t_f} \quad (6.44)$$

Finally, the boundary terms (on RHS) in Eq. (6.44) turn out to be a more general form, whereas Eq. (6.25) applies only to the system of particles model.

For dynamical systems with finite degrees of freedom, using Hamilton's law of varying action does not provide any further information. Hamilton's law has clear advantages when developing the equations of motion of an infinite degree of freedom dynamical system because it not only provides the equations of motion, but also the flexible domain boundary conditions.

Example 6.4: variable cross-sectional axial rod

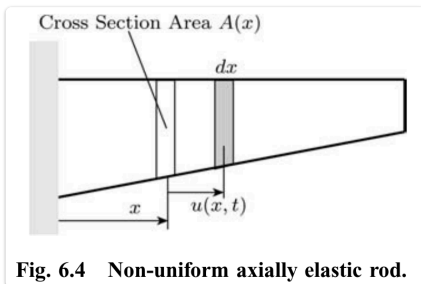


Fig. 6.4 Non-uniform axially elastic rod.

This rod is considered elastic, and neglecting all external forces and internal energy dissipation, this system is conservative. As the elastic simplest constitutive model, for uniaxial stress, the strain energy per unit volume is $\frac{1}{2} E \varepsilon^2$, where $\varepsilon \equiv \frac{\partial u}{\partial x}$ is the strain and E is Young's modulus. Solve for the field function $u(x, t)$ which describe the motion at time t and position x .

For a differential mass $dm = \rho(x)A(x)dx$, the differential kinetic energy is

$$dT = \frac{1}{2} \rho(x) A(x) \left(\frac{\partial u}{\partial t} \right)^2 dx \quad (6.47)$$

The differential potential energy due to strain is

$$dV = \frac{1}{2} E A(x) \varepsilon^2 dx \equiv \frac{1}{2} E A(x) \left(\frac{\partial u(x, t)}{\partial x} \right)^2 dx \quad (6.49)$$

For the entire rod, the total kinetic energy is

$$T = \frac{1}{2} \int_0^L \rho(x) A(x) \left(\frac{\partial u(x, t)}{\partial t} \right)^2 dx \quad (6.48)$$

and the total potential energy is

$$V = \frac{1}{2} \int_0^L E A(x) \left(\frac{\partial u(x, t)}{\partial x} \right)^2 dx \quad (6.50)$$

Substitute into Hamilton's principle and we have:

$$\begin{aligned} \delta \int_{t_0}^{t_f} (T - V) dt &= \int_{t_0}^{t_f} (\delta T - \delta V) dt \\ &= \int_{t_0}^{t_f} \left[\int_0^L \rho A \left(\frac{\partial u}{\partial t} \right) \delta \left(\frac{\partial u}{\partial t} \right) dx - \int_0^L E A(x) \left(\frac{\partial u}{\partial x} \right) \delta \left(\frac{\partial u}{\partial x} \right) dx \right] dt \\ &= \int_0^L \left[\int_{t_0}^{t_f} \rho A \left(\frac{\partial u}{\partial t} \right) \delta \left(\frac{\partial u}{\partial t} \right) dt \right] dx - \int_{t_0}^{t_f} \left[\int_0^L E A(x) \left(\frac{\partial u}{\partial x} \right) \delta \left(\frac{\partial u}{\partial x} \right) dx \right] dt \end{aligned} \quad (6.55)$$

Next, we are going to use integration by parts to unify terms involving $\frac{\partial u(x, t)}{\partial t}$ and $\frac{\partial u(x, t)}{\partial x}$ to terms of $\delta u(x, t)$ only, so that we can then argue δu to be arbitrary and get equations of motion.

With this observation, the two integrations by parts in Eq. (6.55) are

$$\int_{t_0}^{t_f} \rho(x) A(x) \left(\frac{\partial u}{\partial t} \right) \delta \left(\frac{\partial u}{\partial t} \right) dt = \left[\rho(x) A(x) \left(\frac{\partial u}{\partial t} \right) \delta u \right]_{t_0}^{t_f} - \int_{t_0}^{t_f} \rho(x) A(x) \left(\frac{\partial^2 u}{\partial t^2} \right) \delta u dt$$

and

$$\int_0^L E A(x) \left(\frac{\partial u}{\partial x} \right) \delta \left(\frac{\partial u}{\partial x} \right) dx = \left[E A(x) \left(\frac{\partial u}{\partial x} \right) \delta u \right]_0^L - \int_0^L \frac{\partial}{\partial x} \left(E A(x) \left(\frac{\partial u}{\partial x} \right) \right) \delta u dx$$

where we have used $\delta \frac{\partial u}{\partial t} = \frac{\partial}{\partial t} (\delta u)$ and $\delta \frac{\partial u}{\partial x} = \frac{\partial}{\partial x} (\delta u)$.

The conditions $\delta u(x, t_0) = \delta u(x, t_f) = 0$ indicates that the terminal term of the first integrations by parts disappear, i.e.,

$$\left[\rho(x) A(x) \left(\frac{\partial u}{\partial t} \right) \delta u \right]_{t_0}^{t_f} = 0$$

These conditions reflect that we are solving a initial value problem where the initial condition should be given. Regarding why we can also set variation at end point to be zero is because we know that physically there will be only one true solution at the final moment and mathematically we can impose this condition to derive the equations of motion.

At last, substituting everything back to Eq. (6.55) gives:

$$\int_{t_0}^{t_f} \left\{ \int_0^L \left[\frac{\partial}{\partial x} \left(E A(x) \frac{\partial u}{\partial x} \right) - \rho(x) A(x) \frac{\partial^2 u}{\partial t^2} \right] \delta u dx - \left[E A(x) \frac{\partial u}{\partial x} \delta u(x, t) \right]_0^L \right\} dt = 0 \quad (6.58)$$

and we can argue that since $\delta u(x, t)$ is arbitrary, the governing equation of motion is

$$\rho(x) A(x) \frac{\partial^2 u}{\partial t^2} - \frac{\partial}{\partial x} \left(E A(x) \frac{\partial u}{\partial x} \right) = 0 \quad (6.59)$$

together with the boundary conditions

$$E A(0) \frac{\partial u(x, t)}{\partial x} \bigg|_{x=0} \delta u(0, t) = 0 \quad (6.60)$$

$$E A(L) \frac{\partial u(x, t)}{\partial x} \bigg|_{x=L} \delta u(L, t) = 0 \quad (6.61)$$

📌 The results are known widely as the wave equation.

🔗 After-class extension about transversality conditions.

- What would happen if we don't impose the conditions $\delta u(x, t_0) = \delta u(x, t_f) = 0$?
- Why don't we just impose similar conditions to get rid of Eqs. (6.60) and (6.61)?

Hybrid Coordinate Systems

We consider a family of hybrid coordinate systems to be regarded as consisting of a collection of rigid and elastic bodies.

It would be advantageous if analogous results existed for distributed parameter systems, i.e., simple differentiation of energy functions leads to the equations of motion, rather than more tedious integration by parts.

Omitted here and are referred to the textbook and references therein. More intensive knowledge is required if you want to solve a flexible dynamics problem.

Summary

1. Hamilton's principle function $S \equiv \int_{t_0}^{t_f} \mathcal{L} dt$ connects the dynamics to variational calculus.
2. Since Hamilton's principle treat the entire solution as a whole, a functional, all the boundary conditions will appear naturally as transversality conditions. This is a more holistic treatment compared to previous approaches we discussed in this class.
3. Hamilton's principle in this chapter enables a new way to find the solution of a motion directly, but for most cases it still relies on falling back to Lagrange's equations to get the differential equations, which makes it seemingly less powerful.